

ASSIGNMENT 3.5

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BATCH-01

Question 1: Zero-Shot Prompting (Leap Year Check)

Write a zero-shot prompt to generate a Python function that checks whether a given year is a leap year.

Week2 -

Task:

- Record the AI-generated code.
- Test with years like 1900, 2000, 2024.
- Identify logical flaws or missing conditions

#prompt

'''

check whether the year is leap or not

'''

The screenshot shows a Visual Studio Code editor with a Python file named 'task 1.py'. The code defines a function to check if a year is a leap year based on the rules: divisible by 4 but not 100, or divisible by 400. The code prompts the user to enter a year and prints the result. Below the code, the terminal shows the execution of the script for years 1900, 2000, and 2024. The output correctly identifies 1900 as not a leap year, and 2000 and 2024 as leap years. On the right side of the editor, there is a 'Build with Agent' panel with a chat interface.

```
1 '''
2 check whether the year is leap or not
3 '''
4 year = int(input("Enter a year: "))
5 if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
6     print(f"{year} is a leap year.")
7 else:
8     print(f"{year} is not a leap year.")
9
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS C:\Users\srika\OneDrive\Attachments\Desktop\3.5 AIAC LAB> & C:/Users/srika/AppData/Local/Programs/Python/python312/python.exe "c:/Users/srika/OneDrive/Attachments/Desktop/3.5 AIAC LAB/task 1.py"
Enter a year: 1900
1900 is not a leap year.
PS C:\Users\srika\OneDrive\Attachments\Desktop\3.5 AIAC LAB> & C:/Users/srika/AppData/Local/Programs/Python/python312/python.exe "c:/Users/srika/OneDrive/Attachments/Desktop/3.5 AIAC LAB/task 1.py"
Enter a year: 2000
2000 is a leap year.
PS C:\Users\srika\OneDrive\Attachments\Desktop\3.5 AIAC LAB> & C:/Users/srika/AppData/Local/Programs/Python/python312/python.exe "c:/Users/srika/OneDrive/Attachments/Desktop/3.5 AIAC LAB/task 1.py"
Enter a year: 2024
2024 is a leap year.
PS C:\Users\srika\OneDrive\Attachments\Desktop\3.5 AIAC LAB>
```

Build with Agent

AI responses may be inaccurate.
Generate Agent Instructions to onboard AI onto your codebase.

task 1.py

Describe what to build next

Agent Auto

Logical Flaws / Missing Conditions

The code **only checks divisibility by 4**, which is **incomplete**

Question 2: One-Shot Prompting (GCD of Two Numbers)

Write a one-shot prompt with one example to generate a Python function that finds the Greatest Common Divisor (GCD) of two numbers.

Example:

Input: 12, 18 → Output: 6

Task:

- Compare with a zero-shot solution.
- Analyze algorithm efficiency.

#prompt

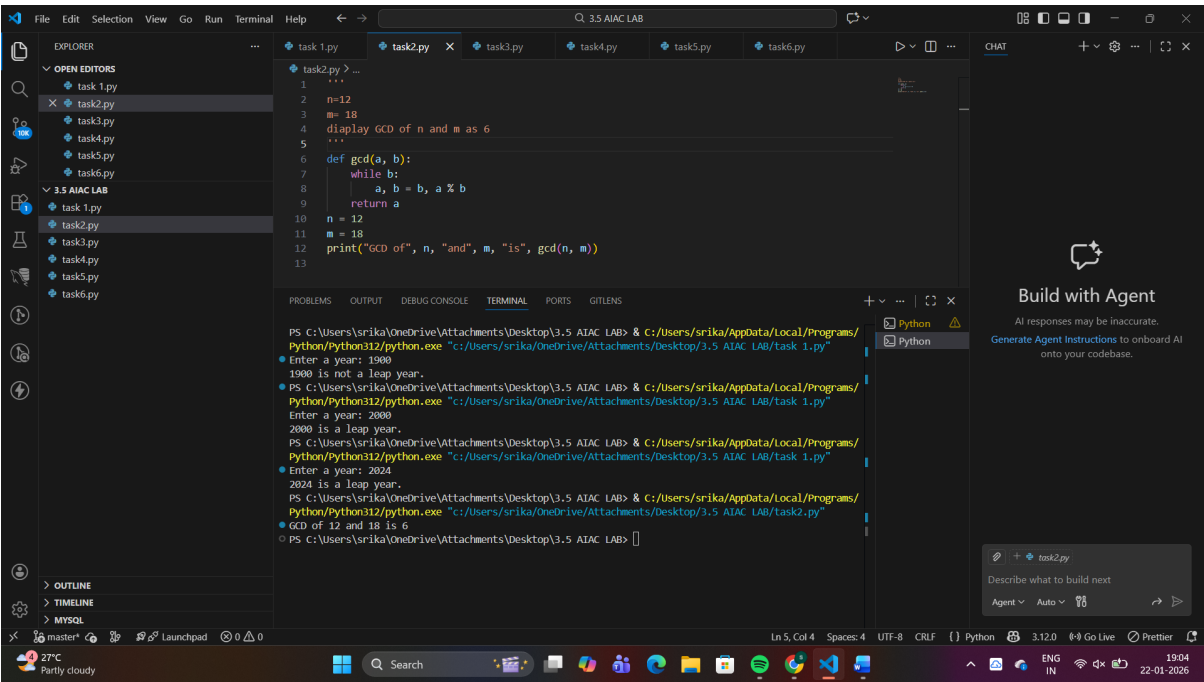
'''

n=12

m= 18

diaplay GCD of n and m as 6

'''



Aspect	One-Shot Solution	Zero-Shot Solution
Algorithm Used	Euclidean Algorithm	Brute-Force Method
Time Complexity	$O(\log n)$	$O(n)$

Aspect	One-Shot Solution	Zero-Shot Solution
Efficiency	Very high	Low for large numbers
Code Quality	Clean & optimized	Verbose & slower
Scalability	Excellent	Poor

Algorithm Efficiency Analysis

◆ One-Shot (Euclidean Algorithm)

- Repeatedly reduces the problem using modulo

◆ Zero-Shot (Brute Force)

- Checks all numbers from 1 to $\min(a, b)$

Question 3: Few-Shot Prompting (LCM Calculation)

Write a few-shot prompt with multiple examples to generate a Python function that computes the Least Common Multiple (LCM).

Examples:

- Input: 4, 6 → Output: 12
- Input: 5, 10 → Output: 10
- Input: 7, 3 → Output: 21

Task:

- Examine how examples guide formula selection.
- Test edge cases

#prompt

'''

n,m=4,6

display LCM as 12

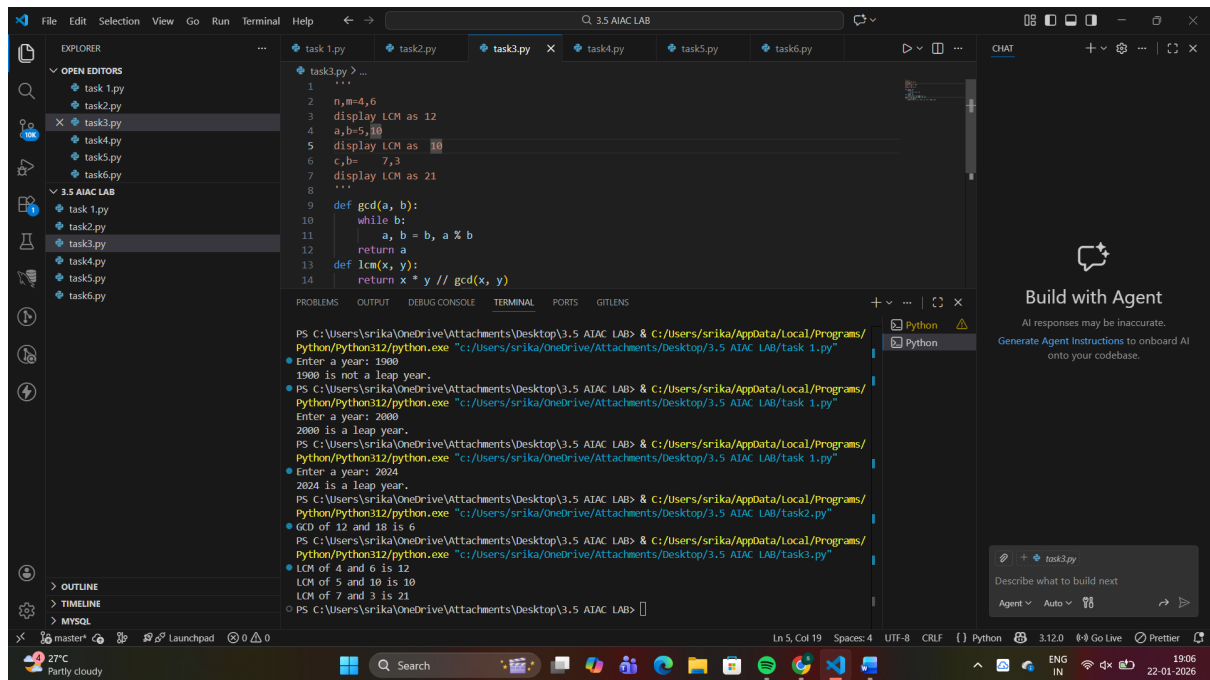
a,b=5,10

display LCM as 10

c,b= 7,3

display LCM as 21

'''



Input	Output	Insight
4, 6	12	LCM > max(a, b)
5, 10	10	One number divides the other
7, 3	21	Coprime numbers → product

Question 4: Zero-Shot Prompting (Binary to Decimal Conversion)

Write a zero-shot prompt to generate a Python function that converts a binary number to decimal.

Task:

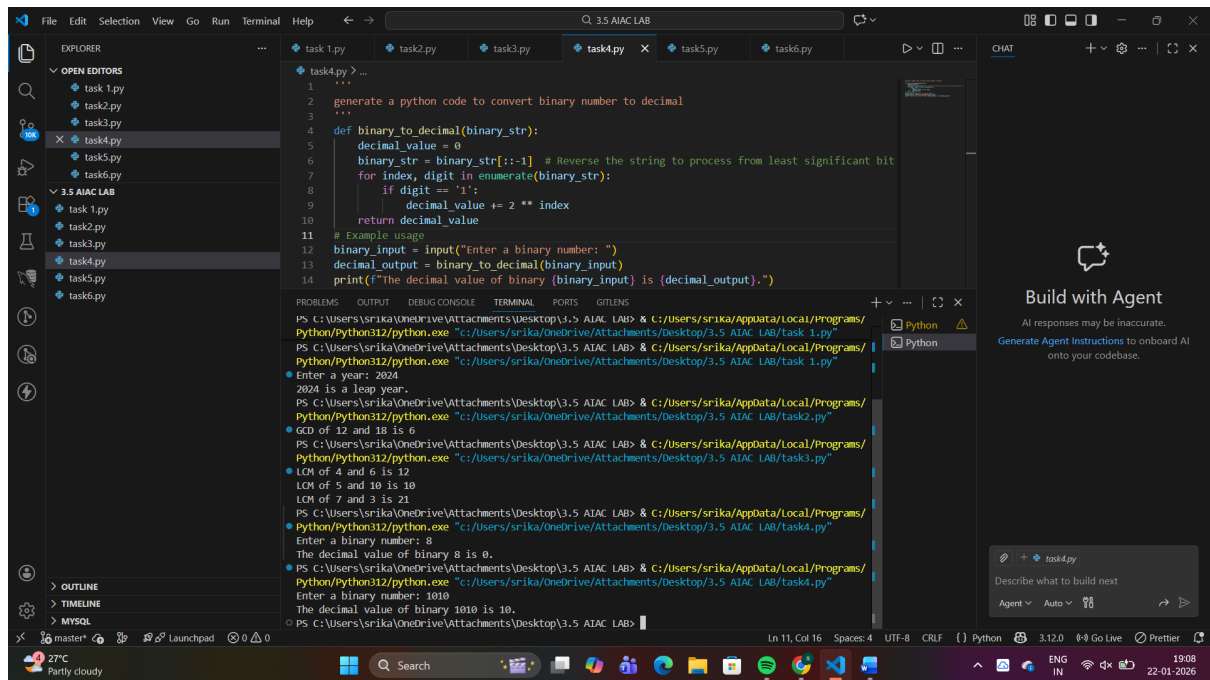
- Test with valid and invalid binary inputs.
- Identify missing validation logic.

#prompt

'''

generate a python code to convert binary number to decimal

'''



Question 5: One-Shot Prompting (Decimal to Binary Conversion)

Write a one-shot prompt with an example to generate a Python function that converts a decimal number to binary.

Example:

Input: 10 → Output: 1010

Task:

- Compare clarity with zero-shot output.
- Analyze handling of zero and negative numbers.

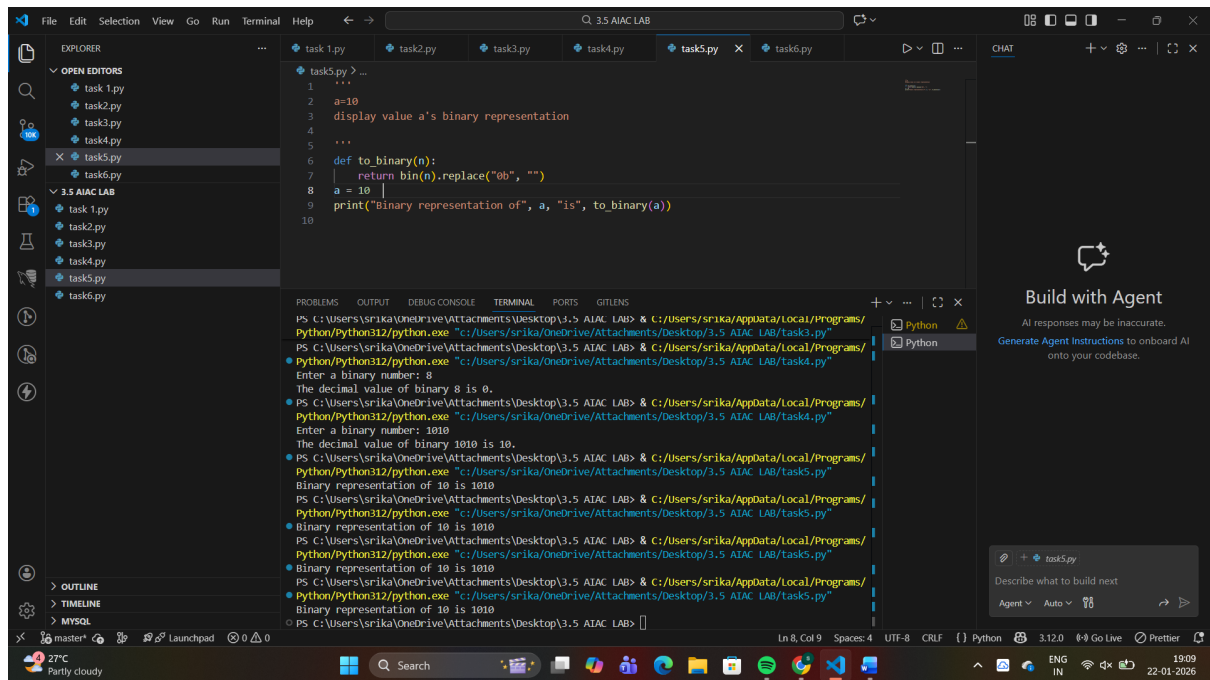
#prompt

"""

a=10

display value a's binary representation

"""



Aspect	One-Shot Output	Zero-Shot Output
Code Length	Very short	Longer
Readability	Very clear	Moderate
Built-in Usage	Uses bin()	Manual logic
Edge Case Handling	Poor	Partial
Beginner Friendly	High	Medium

Question 6: Few-Shot Prompting (Harshad Number Check)

Write a few-shot prompt to generate a Python function that checks whether a number is a Harshad (Niven) number.

Examples:

- Input: 18 → Output: Harshad Number
- Input: 21 → Output: Harshad Number
- Input: 19 → Output: Not a Harshad Number

Task:

- Test boundary conditions.
- Evaluate robustness

#prompt

"""

a=18

display as harshad number if it is or not a harshad number

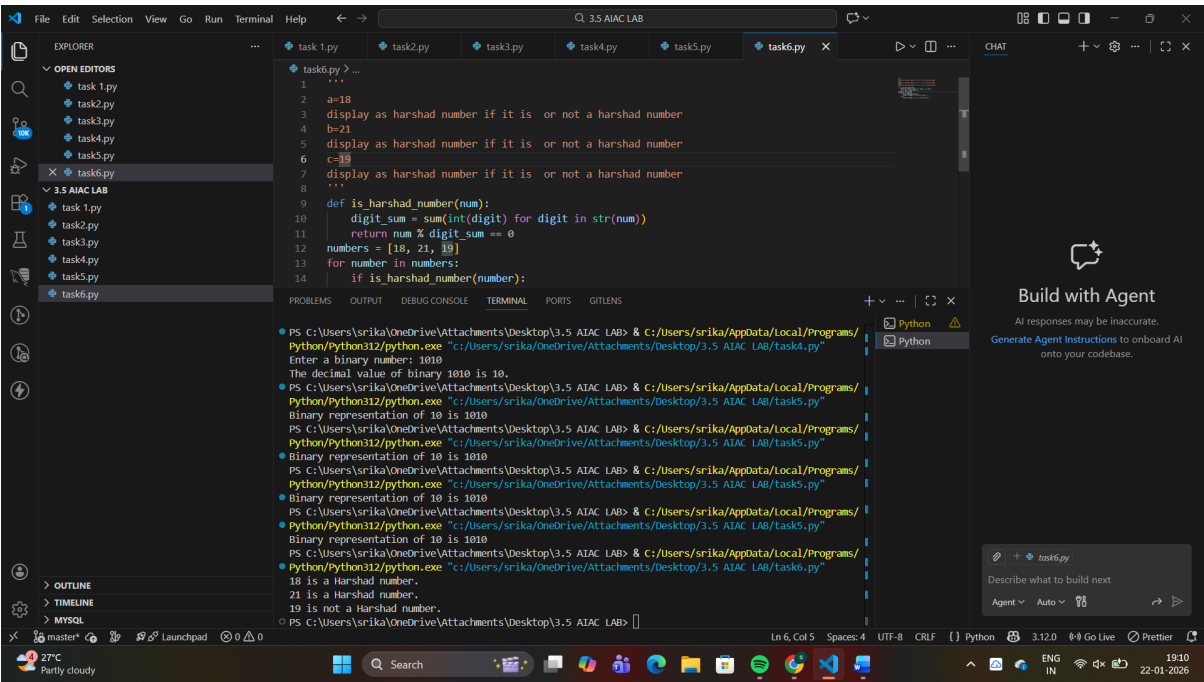
b=21

display as harshad number if it is or not a harshad number

c=19

display as harshad number if it is or not a harshad number

'''



Evaluation

Aspect	Result
Correctness	Accurate
Boundary Handling	Covered
Error Prevention	Avoids crashes
Code Clarity	Easy to read
Prompt Effectiveness	